

Week 4 Report

Gaussian-HMM Evaluation on S&P 500 and Synthetic Data

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CSCD94

1. S&P 500 Evaluation
2. Synthetic Data Evaluation

1. A likelihood-based test was conducted using data from 1990 to 2000 to select the number of regimes.
 - The three-regime model performed best.
2. Different initial parameter values were tested for a two-state Hidden Markov Model (HMM), using a scoring function based on the training data.
 - All initializations converged to the same model.
3. Two-state HMMs were compared using rolling windows of 10, 20, and 60 days.
 - The 60-day rolling window achieved the lowest Bayesian Information Criterion (BIC) and Akaike Information Criterion (AIC).

Regimes	Parameters	Train log-likelihood	AIC	BIC	Best seed	Converged	Iterations
2	7	−3343.031	6700.062	6740.908	20263009	True	61
3	14	−3300.569	6629.139	6710.831	20264010	False	150
4	23	−3275.718	6597.436	6731.645	20265008	False	150

Table 1: Vanilla HMM model comparison by number of regimes

Likelihood Test

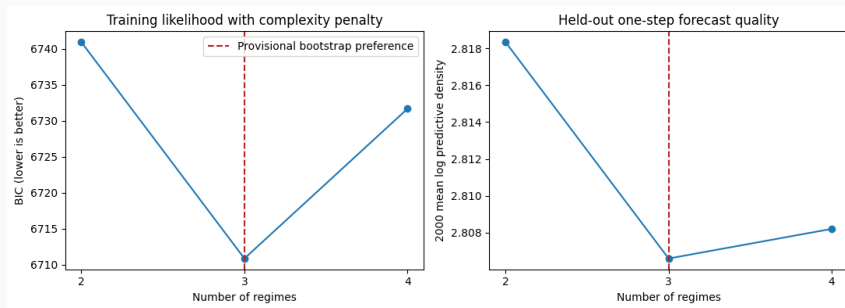


Figure 1: Likelihood Test with bootstrapping test

Two state HMM with different initialization params

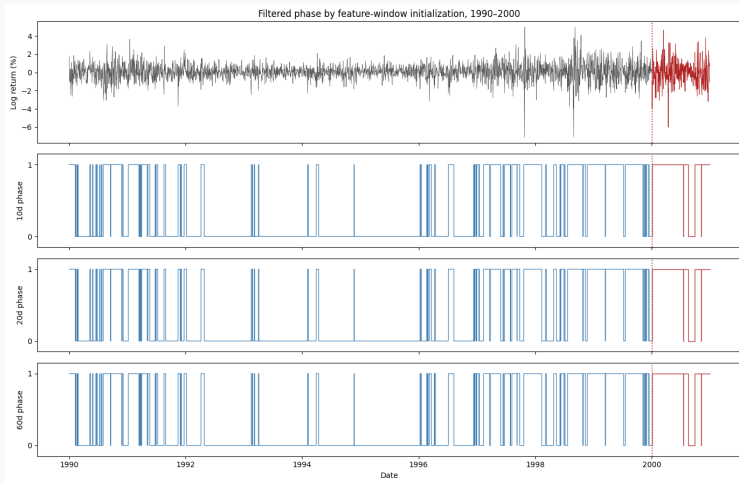


Figure 2: Different initialization param based on the rolling parameters score function

Two state HMM with different rolling params

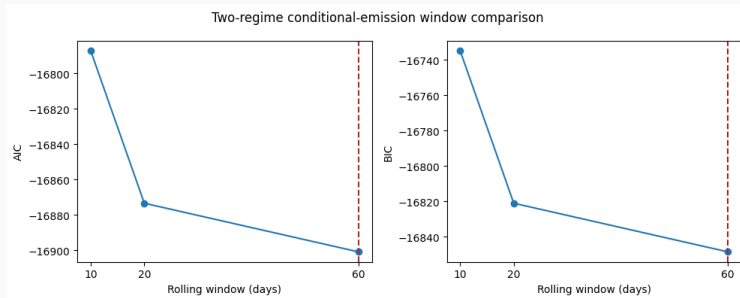


Figure 3: Vanilla HMM with different rolling parameters

Two state HMM with different rolling params

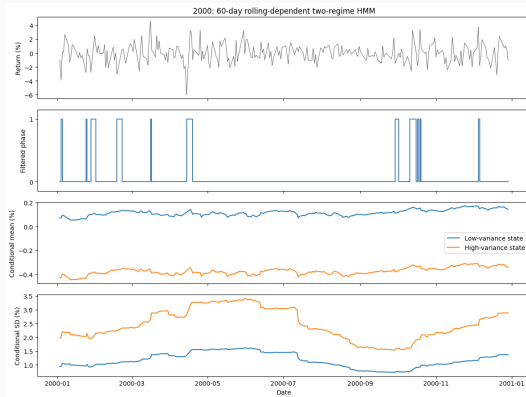


Figure 4: Predicted plot from trained HMM on 1990 to 2000 with rolling 60 mean and volatility

Recall from last weeks, we generate synthetic data to be able to evaluate the model empirically on our objective.

By generating the data ourselves, we know the underlying data generating process, as well as the "hidden" regimes' real labels.

For this experiment, we want to see how the effect of violating several aspects of an assumption of gaussian on the emission can affect its regime recovery capabilities.

These metrics will also serve as a baseline/benchmark for when we implement other emission models.

However, there is still a problem we need to address before evaluation.

Problem: Arbitrary Labels

- HMM hidden-state labels are **arbitrary**.
- The model is not told which state should be called 0, 1, or 2.

True states: 0 1 2

Learned states: 2 0 1

These can represent the **same regimes**, only with different labels.

State labels must be aligned before evaluation
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Solution: Aligning the Learned States

For each learned state, count how often it overlaps with each true state.

	True 0	True 1	True 2
Learned 0	20	450	15
Learned 1	10	25	430
Learned 2	470	15	20

Choose the one-to-one assignment with the largest total overlap:

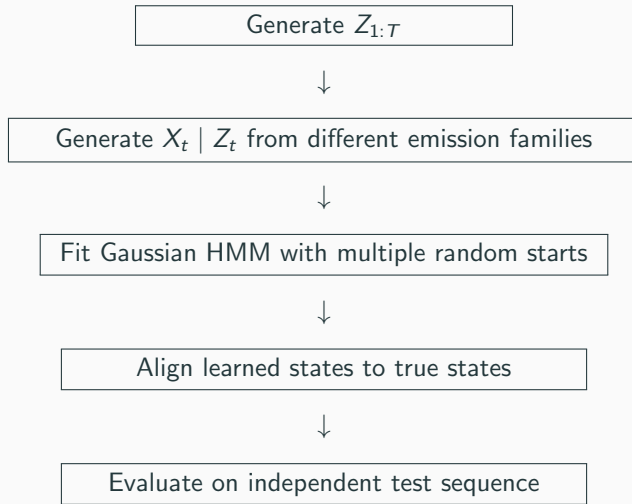
$$0 \rightarrow 1, \quad 1 \rightarrow 2, \quad 2 \rightarrow 0$$

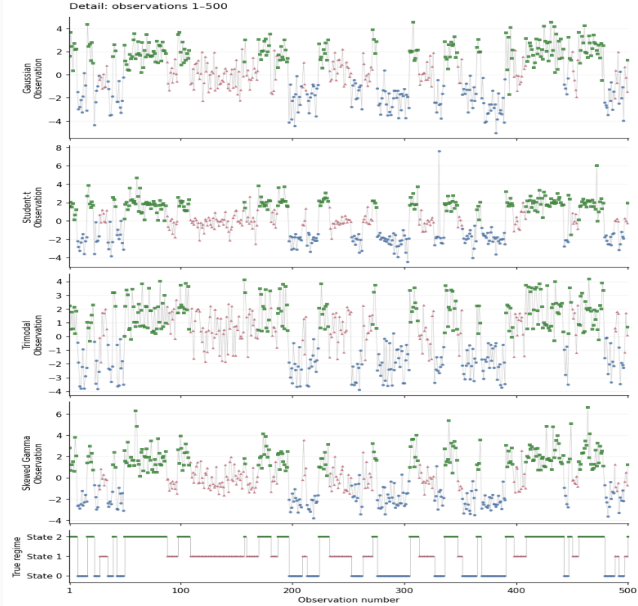
This is an [assignment problem](#).

1. Fit the HMM using only training observations.
2. Use known synthetic training labels to find the state mapping which maximizes the total number of matched observations.
3. Freeze this mapping.
4. Apply it to predictions on the test set.

Solved using `scipy.optimize.linear_sum_assignment` (modified Jonker–Volgenant algorithm).

Experiment Pipeline





Experiment Configuration Summary

- Hidden states: $K = 3$
- State means:

$$\mu = (-2, 0, 2)$$

- State variance:

$$\sigma^2 = 1$$

- Markov chain parameters:

$$\pi = (1/3, 1/3, 1/3), A_{ii} = 0.90, A_{ij} = 0.05$$

- Emissions: Gaussian, Student- t , Gamma, Trimodal, Wide Trimodal
- Student- t : $\nu = 2.5$
- Gamma: shape $\alpha = 4$
- Training sizes:
1,000, 5,000, 20,000
- Test size: 5,000
- 20 replicates datasets, 10 random starts

Synthetic Experiment: Results

Emission	KL	Accuracy	ARI	State Log Loss
Gaussian	0.0003	0.940	0.829	0.157
Skewed Gamma	0.1061	0.933	0.812	0.197
Trimodal	0.2950	0.934	0.815	0.160
Wide Trimodal	41.460432	0.566	0.187	8.652
Student- t	0.7988	0.832	0.709	0.940

Results shown for $N_{\text{train}} = 20,000$ and $N_{\text{test}} = 5,000$ averaged over 20 replicates.

- Gaussian emissions are recovered almost exactly.
- Moderate skewness and multimodality have little effect on state accuracy.
- Wide Trimodal and heavy-tail Student- t 's performance are worse relative to the others, especially for Wide Trimodal.

Best Gaussian KL Approximation

The best Gaussian approximation to a target distribution P under forward KL divergence is

$$\min_{\mu, \sigma^2} D_{\text{KL}} (P \parallel \mathcal{N}(\mu, \sigma^2)) .$$

For the Gaussian family, minimizing forward KL corresponds to moment matching, so the optimal Gaussian has the same mean and variance as P [Kurz et al., 2016]:

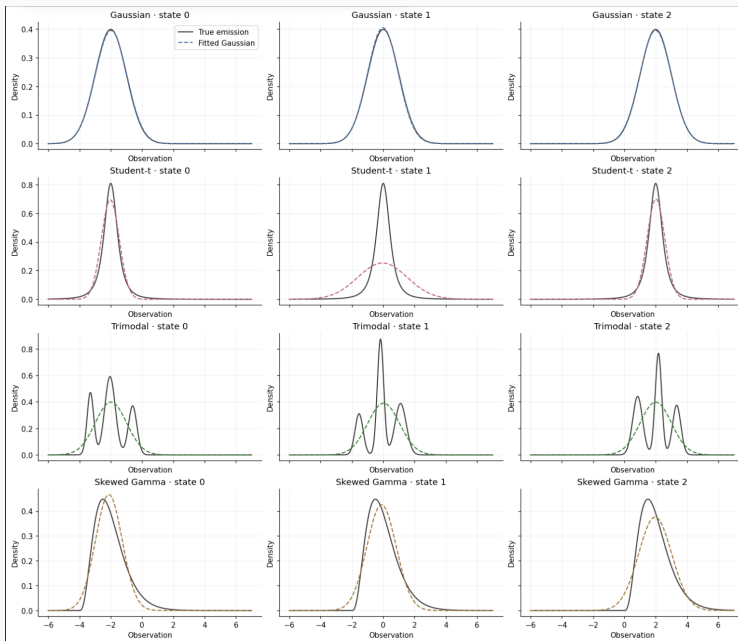
$$\mu^* = \mathbb{E}_P[X], \quad \sigma^{2*} = \text{Var}_P(X).$$

Emission	Fitted KL	Best Gaussian KL	Excess KL
Gaussian	0.0003	0.0000	0.0003
Skewed Gamma	0.1061	0.0887	0.0174
Trimodal	0.2950	0.2946	0.0004
Wide Trimodal	41.4604	1.67237	39.7881
Student- t	0.7988	0.3759	0.4229

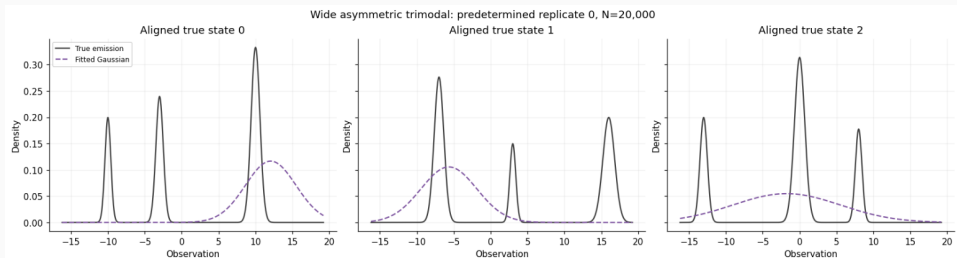
- **Best Gaussian KL:** unavoidable error from approximating the true emission with a Gaussian.
- **Excess KL:** additional error introduced by the fitted HMM.

Effect on Regime Parameter Recovery

Emission	Mean RMSE	Variance RMSE	Transition RMSE
Gaussian	0.014	0.021	0.004
Gamma	0.097	0.187	0.013
Trimodal	0.016	0.034	0.005
Wide Trimodal	8.079	70.160	0.324
Student- t	0.452	5.634	0.137



Wide Trimodal



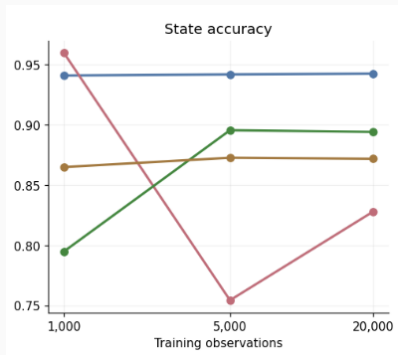
Training Data Size effect on Testing Metrics

— Gaussian

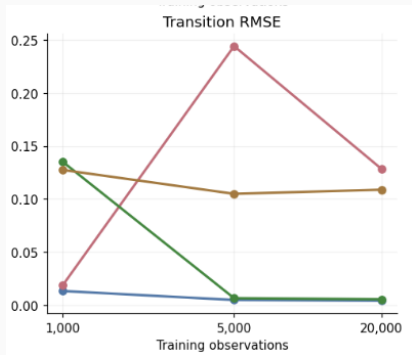
— Student- t

— Trimodal

— Skewed Gamma



Aligned State Accuracy



Transition RMSE

Interpretation

- Increasing training data size past a certain point does not meaningfully improve regime recovery for fitting Gaussian HMMs on non-Gaussian regime conditioned data.
- For skewed Gamma and trimodal emissions, the true density is non-Gaussian, but the model still recovers the underlying regimes well.
- Heavy-tailed Student- t and wide-spread trimodal emissions behave differently: the Gaussian HMM inflates or reorganizes its learned state distributions to explain extreme observations.

Main Finding

The effect of Gaussian emission misspecification depends strongly on **how** the true distribution is non-Gaussian. HMMs with Gaussian emissions is robust to unimodal distributions without fitting its shape, whereas distributions with wide-spread modes and heavy tail behaviour significantly degrades HMM's performance with Gaussian Emissions on regime recovery.

- Implement GMM-HMM
- Evaluate and compare GMM-HMM with Gaussian HMM

Gerhard Kurz, Florian Pfaff, and Uwe D. Hanebeck. Kullback-leibler divergence and moment matching for hyperspherical probability distributions. In *2016 19th International Conference on Information Fusion (FUSION)*, pages 2087–2094, 2016.